

Quiz (Carolina Reaper)

- Q1.** Simplify $\sqrt{64x^3}$, given $x > 0$.
(A) $8x\sqrt{x}$
(B) $4x\sqrt{x}$
(C) $8x^2$
(D) $4x^2$
(E) $2x\sqrt{x}$
- Q2.** A shipping company needs to package boxes for delivery. If the volume of each box is $\sqrt[3]{512}$ cubic feet, how many cubic feet is that?
(A) 8
(B) 16
(C) 64
(D) 512
(E) 729
- Q3.** Express $\sqrt{121}$ in rational exponent form.
(A) $11^{1/2}$
(B) $11^{3/2}$
(C) $11^{2/3}$
(D) 11^1
(E) 11^3
- Q4.** A traffic monitoring system uses $\sqrt{x}$ to model traffic flow. If $x = 900$, what is $\sqrt{x}$?
(A) 20
(B) 25
(C) 30
(D) 15
(E) 45
- Q5.** Simplify $\sqrt{16x^2}$.
(A) $|4x|$
(B) $4x^2$
(C) $\sqrt{4x}$
(D) $8x$
(E) $2x$
- Q6.** A logistics company ships packages with a maximum weight of $\sqrt[3]{1000}$ pounds. What is that weight in pounds?
(A) 5
(B) 8
(C) 10
(D) 20
(E) 25
- Q7.** A transportation system uses the formula $d = rt$ to calculate distance traveled, where r is the rate of travel in miles per hour and t is time in hours. If the rate is $\sqrt{16t}$, what is the distance traveled in 2 hours?
(A) 4
(B) 8
(C) 16
(D) 32
(E) 64
- Q8.** Convert $\sqrt[3]{27}$ to rational exponent form.
(A) $3^{1/3}$
(B) $3^{2/3}$
(C) 3^3
(D) 3^1
(E) 3^2

Open Ended Questions (FRQ)

- Q9.** A delivery company has a truck that travels at a rate of $\sqrt{x}$ miles per hour. If the truck travels 12 miles in 2 hours, what is the value of x ? Show your work and explain your reasoning.
- Q10.** A traffic engineer uses the formula $v = \sqrt{2gd}$ to calculate the velocity of traffic flow, where v is the velocity, g is the acceleration due to gravity, and d is the distance traveled. If the distance traveled is $\sqrt[3]{64}$ miles, what is the velocity of the traffic flow? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Ensure students understand the concept of radicals and rational exponents.
- Tip 2: Encourage students to use real-world examples to illustrate their understanding of the concepts.
- Tip 3: Remind students to simplify their answers and show their work.